

Yongji Wang

Science and Strategic Initiatives, Google DeepMind
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EDUCATION

Massachusetts Institute of Technology Ph.D. in Civil and Environmental Engineering – Thesis: “Fundamentals in unsteady fluid fragmentation from drop impact”	Cambridge, MA 09/2014 –01/2021
University of Cambridge M.A.St in Applied Mathematics (Part III Tripos) – Thesis: “The transmission of shock waves in curved tubes”	Cambridge, UK 09/2012 –07/2013
University of Hong Kong B.S. in Mechanical Engineering – Thesis: “Nonlinear shallow water waves on running water”	Hong Kong 09/2010 –06/2012
Shanghai Jiao Tong University B.S. in Mechanical Engineering (dual degree with the University of Hong Kong) – Dual degree program with Hong Kong University	Shanghai, CN 09/2008 –06/2010

RESEARCH EXPERIENCE

Google DeepMind Research Scientist in the <i>Science and Strategic Initiatives</i> • Developing machine-learning based solver for nonlinear equations	New York, NY 10/2025 - present
New York University Postdoctoral Fellow / Instructor in the <i>Courant Institute</i> Mentor: Tristan Buckmaster • Discovery of unstable singularities to nonlinear fluid equations via PINNs – Developed new computational approach of embedding mathematical insights into neural network training in discovering elusive unstable solutions to complex nonlinear differential equations. – Discovered, for the first time, series of unstable blow-up solutions to challenging nonlinear fluids equations and unstable solitons to Nonlinear Schrödinger equation. The work is under review at <i>Nature</i> .	New York, NY 10/2023 - 10/2025
Visiting postdoctoral associate at Stanford University Co-mentor: Ching-Yao Lai • Investigation of viscoelastic properties of Antarctica Ice Shelves via deep learning – Developed a novel, open-source algorithm that leverages satellite imagery and physics-informed machine learning to infer flow law of Antarctic ice shelves. – Discovered the anisotropic property of Antarctic ice shelves using remote observations and the self-developed machine learning algorithm. The work was published in <i>Science</i> .	10/2023 - 10/2024
Princeton University Postdoctoral research associate in the <i>Department of Geosciences</i> Mentor: Ching-Yao Lai • Data-driven discovery of flow laws of ice shelves – Coupled observations with physics model to reveal the rheology of Antarctic Ice Shelves.	Princeton, NJ 02/2021 - 09/2023

- Leveraged the remote-sensing observations of ice velocity and thickness and physics-informed neural network to infer two-dimensional effective ice viscosity, strain rate and deviatoric stress of Antarctic Ice Shelves.
- Discovered the substantial difference of ice-shelf rheology between the compression zone, where the ice rheology closely obeys power laws, and extension zones, where ice performs complex rheological behaviors.

Adjunct postdoctoral associate in the *Department of Mathematics* Co-mentor: Tristan Buckmaster

- Discovery of self-similar blow-up solution via physics-informed neural network 08/2021 - 08/2023
- Established new numerical approach, leveraging physics-informed deep learning and multi-stage training scheme to search for self-similar blow-up solutions to nonlinear fluid equations with high accuracy and efficiency.
- Discovered, for the first time, the self-similar blow-up solution for the 3-dimensional axisymmetric Euler equations in the presence of cylindrical boundary, which sheds new light to the field of mathematical fluid dynamics.
- Discovered, for the first time, the *unstable* self-similar blow-up solutions for the Córdoba-Córdoba-Fontelos (CCF) equations, a milestone towards the solution to the Millennium Prize Navier-Stokes problem.

Massachusetts Institute of Technology

Cambridge, MA

Ph.D student in the *Civil and Environmental Engineering* Advisor: Lydia Bourouiba

- Persistence of respiratory cloud in shaping disease transmissions 03/2020 - 09/2020
- Innovated integrated experimental platform, combining Schlieren Imaging, Interferometry Particle Imaging, and Particle Tracking Imaging to capture both gas and droplet phases of human respiratory clouds.
- Developed hybrid particle tracking algorithms that link Interferometry Particle Imaging with Particle Tracking Velocimetry to measure both size and speed distributions of droplets in respiratory clouds with high precision.
- Fundamentals in unsteady fluid fragmentation on drop impact 09/2015 - 01/2021
- Innovated fully-automatized integrated platform that enables reproduction of complex fragmentation experiments with high efficiency and accuracy, combined with systematic flow visualization and measurement techniques.
- Developed multi-step Advanced Image-Processing algorithms to capture spatio-temporal evolution of key physical quantities of unsteady fragmentation with unprecedented accuracy.
- Established a theoretical framework to elucidate fundamental physics underlying unsteady fragmentation that enables the prediction of both size and speed distributions of droplets generated during fragmentation.
- Drop-on-drop interactions on rigid surfaces 09/2014 - 01/2016
- Discovered four types of interactions upon drop impact in the vicinity of another sessile drop on the surface.
- Rationalized and predicted the boundaries between different types of drop-on-drop interactions.

Massachusetts Institute of Technology

Cambridge, MA

Principal graduate researcher at *change:WATER Labs* (Clean tech. startup)

- Dignified sanitation solutions for non-sewered communities 09/2015 - 06/2018
- Invented low-price, compact, standalone and dignified non-sewered toilet, made of specific-designed membrane that enables rapidly shrinking of urban sewage accumulations by evaporation.
- Designed precise and well-controlled fluid dynamics experiments for measuring the evaporation rate of the toilet under different environmental conditions, including humidity, temperature and pressure.
- Developed advanced data-processing algorithms that enable systematic analysis of measured experimental data for validating and improving toilet performance.
- Configured the ventilation system and internal structure of the non-sewered toilet based on systematic tests to maximize its efficiency to shrink liquid sewage by evaporation.

University of Hong Kong

Hong Kong

Research assistant at the *Aerodynamics and Acoustics Laboratory* Advisor: Lixi Huang

- High-order wave propagation and noise attenuation control in cylindrical ducts 07/2013 - 07/2014
- Innovated sophisticated experimental setups to mimic the fan noise of aircraft engine.
- Wrote computational program for simulating superposition of acoustic pressure field from point sources with different phase angles and amplitudes at any positions in cylindrical tubes.
- Introduced new approach of producing rotating sound in a cylindrical tube with single high-order azimuthal mode and designed the duct lining made of micro-perforated panels for high-frequency noise attenuation.

University of Cambridge

Cambridge, UK

Graduate researcher at the *Waves group* of DAMTP Advisor: Edward J. Brambley

- Transmission of shock waves in curved tubes 12/2012 - 05/2013
- Researched Whitham's geometrical shock dynamics.
- Implemented a simple method to derive the geometrical and dynamical relations between the cross-sectional area of the ray tube and the velocity of the shock front.
- Wrote computational program for simulating the transmission of shock waves in a rectangular tube based on Whitham's geometrical shock dynamics.

University of Hong Kong

Hong Kong

Undergraduate researcher in the department of *Mechanical Engineering* Advisor: Kwok Wing Chow

- Nonlinear shallow water waves on running water 12/2012 - 05/2013
- Studied asymptotic expansion and perturbation theory.
- Mastered the way of deriving the Korteweg-de Vries (KdV) and Kadomtsev-Petviashvili (KP)-type II equations in the case of small-amplitude perturbation and no mean flow.
- Investigated the problem of uniform flow or shear flow, with focus on the conditions of production of lump solitary wave production at the surface, and its compatibility with the first kind of KP equations.

PUBLICATIONS

Under Review

Y, Wang, T. Leger, C. Y. Lai and T. Buckmaster “Resolving sharp gradients of unstable singularities to machine precision ”

Y, Wang, M. Bennani, J. Martens *et al.* “Discovery of unstable singularities”

J. Rines, C. Y. Lai and **Y, Wang**, “Theoretical analysis of stress perturbations from a partially-lubricated viscous gravity current”.

Journal Articles

1. **Y, Wang**, C. Y. Lai, D. J. Prior and C. Breen-Cowen, “Deep learning the flow laws of Antarctic Ice Shelves,” *Science*, **387**,1219-1224, (2025)
2. **Y, Wang**, C. Y. Lai, “DIFFICE-jax: Differentiable neural-network solver for data assimilation of ice shelves in JAX,” *Journal of Open Source Software*, (2025)

3. N. B. Coffey, C. Y. Lai, **Y. Wang**, W. R. Buck, T. Surawy-Stepney, A. E. Hogg., “Theoretical stability of ice shelf basal crevasses with a vertical temperature profile,” *Journal of Glaciology*, 1-22, (2024).
4. **Y. Wang**, and C. Y. Lai, “Multi-stage Neural Networks: Function Approximator of Machine Precision,” *Journal of Computational Physics*, **504**, 112865 (2024).
5. **Y. Wang**, C. Y. Lai, J. Gómez-Serrano, and T. Buckmaster. “Asymptotic self-similar blow up profile for 3-D axisymmetric Euler equation via neural networks,” *Physical Review Letters*, **130**, 244002 (2023)
6. **Y. Wang** and L. Bourouiba, “Non-Galilean Taylor-Culick law governs sheet dynamics in unsteady fragmentation,” *Journal of Fluid Mechanics*, **969**, A19 (2023)
7. **Y. Wang** and L. Bourouiba, “Mass, momentum and energy partitioning of unsteady fragmentation,” *Journal of Fluid Mechanics*, **935**, A29 (2022)
8. **Y. Wang** and L. Bourouiba, “Growth and breakup of ligaments in unsteady fragmentation,” *Journal of Fluid Mechanics*, **910**, A39 (2021)
9. **Y. Wang** and L. Bourouiba, “Unsteady sheet fragmentation: droplet sizes and speeds,” *Journal of Fluid Mechanics*, **848**, 946-967 (2018)
10. **Y. Wang**, R. Dandekar, N. Bustos, S. Poulain and L. Bourouiba, “Universal rim thickness in unsteady sheet fragmentation”, *Physical Review Letters*, **120**, 204503 (2018)
11. **Y. Wang** and L. Bourouiba, “Non-isolated drop impacts on surfaces,” *Journal of Fluid Mechanics*, **835**, 24-44 (2018)
12. **Y. Wang** and L. Bourouiba, “Drop impact on small surfaces: thickness and velocity profiles of the expanding sheet in the air,” *Journal of Fluid Mechanics*, **814**, 510-534 (2017)

Conference Papers

1. E. Y. Wang, **Y. Wang**, C. Y. Lai, “ Training Dynamics of Convolutional Neural Networks for Learning the Derivative Operator,” *Neurips 2024 SciForDL Workshop*, (2024).
2. J. Ng, **Y. Wang**, C. Y. Lai, “Spectrum-informed multistage neural network: Multiscale function approximator of machine precision,” *ICML 2024 AI4Science Workshop*, (2024).
3. C. Cowen-Breen, **Y. Wang**, C. Y. Lai, “Euler operators for mis-specified physics-informed neural networks,” *ICML 2024 AI4Science Workshop*, (2024).

Google Scholar link: [Yongji Wang](#)

TEACHING EXPERIENCE

Princeton University

- Guest lecturer of *Summer School on Fluids and Computer Assisted Proof* Summer 2025
Graduate-level course Class size: 20-30 students
- Provided a detailed survey of state-of-the-art PINN techniques on solving partial differential equations (PDEs).
- Provided a live, guided implementation of advanced PINNs techniques on solve PDEs to solidify comprehension and equip students with the practical skills of employing PINNs in their own research.

- Guest lecturer of *Deep learning in Geophysical Fluid Dynamics (AOS551)* Fall 2021 & 2022
- Graduate-level course Class size: 20-30 students
- Taught students two lectures on fundamental principles and techniques of physics-informed neural networks.
- Prepared code examples for students to study TensorFlow and learn to write basic algorithms using physics-informed neural network.

Massachusetts Institute of Technology

- Teaching Assistant of *Nonlinear Dynamics and Turbulence (1.686/18.358/2.033)* Spring 2017 & 2020
Graduate-level course Class size: 10-15 students Twice
- Held and led office hours, tutorials, and graded problem sets.
- Assisted students to understand lecture materials and illustrate problem sets.
- Teaching Assistant of *Fluids and Diseases (1.063/1.631/2.250/HST.537)* Spring 2017 – 2019
Graduate-level course Class size: 10-15 students Three times
- Improved teaching materials including problem sets and their solutions.
- Prepared equipment for in-class demonstration.
- Taught students recitations and lectures on fundamental fluid dynamics.
- Created teaching demonstration GUIs to illustrate how spatial information affects disease transmission.
- Teaching Assistant of *Nonlinear Dynamics: Continuum Systems (1.062/18.354)* Spring 2016 & 2018
Graduate-level course Class size: 10-15 students Twice
- Improved teaching materials including problem sets and their solutions.
- Prepared equipment of the lab session for students to conduct experiments.
- Taught recitations of the course for students to prepare quizzes and mid-term exams.
- Assistant of *Innovation and Technology in Agriculture and the Environment* Summer 2017
MIT Professional Education Short Program Class size: 10-15 participants
- Prepared equipment and notes for the laboratory demonstration.

MENTORING EXPERIENCE

New York University

- George Lu Ph.D student 06/2024 - present
Topic: Inversion of rheology of Antarctica Ice Sheet from Radar measurement
- Erik Wang Undergraduate student 06/2024 - 09/2024
Topic: Training Dynamics of CNNs for Learning the Derivative Operator
- published a conference paper in the NeuIPS (2024) Workshop: SciForDL
- Joshua Rines Ph.D student 05/2023 - 06/2024
Topic: Stress perturbation of the Greenland Ice Sheet in response to its basal lubrication
- submitted the manuscript of the research to the Journal of Fluid Mechanics
- Charlie Cowen-Breen Ph.D student 01/2024 - 07/2024
Topic: Over-constraint and cheating effect of physics-informed neural networks (PINNs)
- published a conference paper in the ICML (2024) Workshop: AI4Science

- Jakin Ng Undergraduate student 07/2023 - 05/2024
Topic: Specturm information multi-stage neural networks for machine precision
– published a conference paper in the ICML (2024) Workshop: AI4Science

Princeton University

- Evans Chandran Undergraduate student 01/2023 - 05/2023
Junior Project (JP): Investigating glacial ice flow regimes via physics-informed neural networks
- Charlie Cowen-Breen Undergraduate student 02/2021 - 05/2022
Thesis: Investigating glacial ice flow regimes via physics-informed neural networks
– Received the Sigma Xi Book Award in the Math Department for his senior thesis research
– Received the Peter A. Greenberg '77 Prize in the Math department for junior undergraduates

Massachusetts Institute of Technology

- Xinyu Si Master student 05/2019 - 12/2019
Project: Developing light-absorption method for the measurement of interfacial phenomena
– Submitted a paper to the *Experiments in Fluids*
- Tyler J. Pleasant Undergraduate student 01/2016 - 05/2016
Project: Developing image-processing algorithm to analyze drop fragmentation processes
- Budmonde Duinkhar Undergraduate student 11/2014 - 02/2015
Project: Quantifying the fragmentation phenomenon of drop-on-drop interactions

PRESENTATIONS

Invited Talks

- “*Deep learning the physical properties of Antarctic ice shelves*” AOS colloquium, Courant Institute of Mathematical Science, New York University, New York, NY (April 2025)
- “*Unraveling the physical properties of Antarctic ice: A data-driven approach*,” Department colloquium of EPS, Harvard University, Cambridge, MA (Mar. 2025)
- “*Recent Progress in Physics-Informed Neural Networks for Self-Similar Blow-Up Solutions*,” PDE Math seminar, University of Minnesota Twin Cities, Minneapolis, MN (Oct. 2024)
- “*Multi-stage neural networks achieving machine precision*,” Monthly seminar (virtual) at ANSYS Co. (Aug. 2024)
- “*Multi-stage neural networks achieving machine precision*,” Department Seminar of Mathematical Science, Korea Advanced Institute of Science & Technology, Daejeon, Korean (May 2024)
- “*Multi-stage neural networks for discovering self-similar blow-up solutions*,” FRG conference of PDEs of incompressible fluid flows, computer assisted proofs and nueral networks, Minneapolis, MN (April 2024)
- “*Application of fundamental fluid mechanics in disease transmissions*,” Feilan Environmental and Health Seminar, Chinese Society for Environmental Science, Beijing, China (Oct. 2023)

- “*Physics informed neural networks for self-similar blow-up solutions*,” PDE & Applied Math Seminar, University of Maryland, College Park, MD (Nov. 2022)
- “*Physics informed neural networks for blow-up solutions of Euler equations*,” PDE & Applied Math Seminar, University of California, Davis, CA (Oct. 2022)
- “*Physics informed neural networks for self-similar blow-up solutions*,” Scientific Machine Learning Webinar, Carnegie Mellon University, Pittsburgh, PA (Oct. 2022)
- “*Physics-informed neural network for fluid mechanics*,” Challenges and Prospects of ML for the Physical Sciences, Simons Foundation, New York, NY (June 2022)
- “*Physics-informed neural network for blow-up solutions of Euler equation*,” Mathematics, Physics & Machine Learning webinar, Lisbon, Portugal (May 2022)
- “*Physics-informed neural network for self-similar blow-up solutions*,” PDE and Applications Seminar, Uppsala, Sweden (May 2022)

Seminar Talks

- “*Discovering self-similar blow-up solutions using physics-informed neural networks*,” M2lines ML meeting, New York University, NY (May 2024)
- “*Assessing ice rheology using physics-informed deep learning*,” West Antarctic Ice Sheet Workshop, University of Minnesota, MN (Sep. 2023)
- “*Discovering ice-shelf rheology via physics-informed neural network*,” GEO Climate Seminar, Princeton University, NJ (Oct. 2022)
- “*Inferring flow law of ice shelves using physics-informed neural network*,” Soft Matter Coffee Hour, Princeton University, NJ (April 2022)

Conference Talks

- **Y. Wang** and C. Y. Lai “*Study of well-posedness in the inverse problem of inferring ice-shelf viscosity*,” 77th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Salt Lake City, UT (November 2024)
- J. Rines, **Y. Wang** and C. Y. Lai “*Theoretical analysis of stress perturbations from a partially-lubricated viscous gravity current*,” 77th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Salt Lake City, UT (November 2024)
- **Y. Wang**, C. Cowen-Breen, and C. Y. Lai “*Assessing ice rheology using physics-informed deep learning*,” American Geophysical Union Fall Meeting, San Francisco, CA (December 2023)
- **Y. Wang**, C. Y. Lai, J. Gómez-Serrano, and T. Buckmaster “*Discovering self-similar blow-up solutions using physics-informed neural networks*,” 76th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Washington DC (November 2023)
- J. Rines, **Y. Wang** and C. Y. Lai “*Viscous response of ice sheet dynamics to water lubrication at the bed*,” 76th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Washington DC (November 2023)

- **Y. Wang** and C. Y. Lai “*Accessing ice rheology using physics-informed deep learning*,” EGU General Assembly Conference, Vienna, Austria (April 2023)
- N. Coffey, C. Y. Lai, and **Y. Wang** and C. Y. Lai “*Rift Initiation via Unstable Basal Crevasses*,” EGU General Assembly Conference, Vienna, Austria (April 2023)
- **Y. Wang**, C. Cowen-Breen, and C. Y. Lai “*Discovery of ice shelf rheology via physics-informed neural network*,” American Geophysical Union Fall Meeting, Chicago, IL (December 2022)
- J. Rines, **Y. Wang** and C. Y. Lai “*Lake Drainage Cascades Induced by Basal Lubrication*,” American Geophysical Union Fall Meeting, Chicago, IL (December 2022)
- C. Y. Lai, **Y. Wang** and Y. Iwasaki “*Bridging Antarctic observations and models with deep learning*,” American Geophysical Union Fall Meeting, Chicago, IL (December 2022)
- N. Coffey, C. Y. Lai, **Y. Wang** and F. Ishraque “*Ice Shelf Vertical Temperature Profile can Substantially Impact Basal Crevasse Stability*,” American Geophysical Union Fall Meeting, Chicago, IL (December 2022)
- **Y. Wang**, C. Cowen-Breen, and C. Y. Lai “*Discovery of ice shelf rheology via physics-informed neural network*,” 75th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Indianapolis, IN (November 2022)
- **Y. Wang**, E. Pomerantz, C. Cowen-Breen, R. Chou and C. Y. Lai “*Inferring flow law of ice shelves using physics-informed neural network*,” 74th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Phoenix, AZ (November 2021)
- **Y. Wang** and L. Bourouiba “*Unsteady fragmentation upon drop impact: Sheet dynamics*,” 72th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Seattle, WA (November 2019)
- L. Bourouiba and **Y. Wang** “*Unsteady fragmentation upon drop impact: prediction of droplet size and speed distributions*,” 72th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Seattle, WA (November 2019)
- **Y. Wang** and Bourouiba “*Pathogen transmission via droplets during unsteady fragmentation*,” 1st Fluid and Health Conference, Cargèse, Corsica (August 2019)
- **Y. Wang** and L. Bourouiba “*Ligament dynamics along unsteady rims*,” 71th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Atlanta, GA (November 2018)
- L. Bourouiba and **Y. Wang** “*Rim thickness evolution in unsteady sheet fragmentation*,” 71th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Atlanta, GA (November 2018)
- **Y. Wang** and L. Bourouiba “*Rim destabilization in unsteady sheet expansion in the air from drop impact on small surfaces*,” 70th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Denver, CO (November 2017)
- L. Bourouiba and **Y. Wang** “*Role of unsteadiness and local dynamics in the selection of secondary droplet sizes in drop impact fragmentation*,” 70th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Denver, CO (November 2017)

- **Y. Wang** and L. Bourouiba “*Unified thickness profile of radially expanding sheets in the air,*” 69th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Portland, OR (November 2016)
- L. Bourouiba and **Y. Wang** “*Binary drop interaction on surfaces: onset and bounding ligaments of Crescent-Moon fragmentation,*” 69th Annual Meeting of the American Physical Society, Division of Fluid Dynamics, Portland, OR (November 2016)

Competition and Pitch Talks

- **Y. Wang**, C. Smith, H. Elasaad and Y. Cao “*Dignified sanitation solutions for non-sewered communities*” (Third Prize and Audience-Choice Award), MIT \$100k Entrepreneurship Competition, Cambridge, MA (May 2017)
- **Y. Wang**, C. Smith, H. Elasaad and Y. Cao “*Dignified sanitation solutions for non-sewered communities*” (Finalist), Hultz Prize regional competition, Boston, MA (March 2017)
- **Y. Wang** and L. Bourouiba “*Drop fragmentation in shaping disease transmissions*” (Winner), Research Speed Dating Event, Department of Civil and Environmental Engineering, MIT (February 2017)
- **Y. Wang**, C. Smith, H. Elasaad and Y. Cao “*Dignified sanitation solutions for non-sewered communities*” (Winner), Hult Prize competition@MIT, Cambridge, MA (November 2016)

Poster Presentations

- **Y. Wang**, C. Y. Lai, D. Prior and C. Cowen-Breen “*Deep learning the flow law of Antarctic ice shelves,*” American Geophysical Union Fall Meeting, Washington DC, (December 2024)
- **Y. Wang** and C. Y. Lai “*DIFFICE-jax: DIFFerentiable neural-network solver for data assimilation of ICE shelves written in JAX,*” American Geophysical Union Fall Meeting, Washington DC (December 2024)
- **Y. Wang** and C. Y. Lai “*Inferring viscosity of ice shelves using physics-informed neural network,*” Ice-Ocean Interactions on Icy Moons in the Solar System Workshop, Princeton Center for Theoretical Science, Princeton, NJ (April 2022)
- **Y. Wang** and L. Bourouiba “*Fundamentals in unsteady fluid fragmentation,*” Research Speed Dating Event, Department of Civil and Environmental Engineering, MIT (February 2020)
- **Y. Wang** and L. Bourouiba “*Fundamentals in unsteady fluid fragmentation with applications to disease transmissions,*” Industrial Group Symposium of Institute for Medical Engineering and Science, MIT (October 2019)

HONORS AND AWARDS

Academic Awards

- John S. Hennessy OGE Fellowship 2017
Competitive institutional fellowship (one per year) awarded to graduate students with strong academic and research performance by the Office of Graduate Education, Massachusetts Institute of Technology.

- MIT GSC Travel Grant
2017 & 2019

Competitive institutional travel grant awarded by the Graduate Student Council, Massachusetts Institute of Technology, in support of travel expenses for graduate students who present talks in academic-related conferences.
- Best Talk Award of MIT CEE Research Speed Dating Event
2017

The first prize of the graduate student presentation competition at the Research Speed Dating Event hosted by the Department of Civil and Environmental Engineering, Massachusetts Institute of Technology.
- Berger Fellowship
2015

Honorific fellowship awarded to newly-enrolled graduate students with strong academic and research background by the Department of Civil and Environmental Engineering, Massachusetts Institute of Technology
- Cambridge Darwin College Prize
2013

Honorific prize awarded to the students in Darwin College who achieve Distinction for their degrees by Darwin College, the University of Cambridge.
- Dean's Honor List of the Faculty of Engineering, HKU
2011 & 2012

Highest academic prize awarded to the top 5% undergraduate students with the best academic performance by the Faculty of Engineering, the University of Hong Kong.
- CEDES Scholarship
2008-2012

Competitive institutional 4-year scholarship (two per year) awarded to undergraduate students of Shanghai Jiao Tong University with strong academic performance by the CEDES Corporation.
- Merit Student of Shanghai Jiao Tong University
2010

Highest individual prize awarded to undergraduate students with the best comprehensive qualities including academic, research and social activities by Shanghai Jiao Tong University.
- Top Ten learning pacesetter of the Faculty of Engineering, SJTU
2010

Highest academic prize awarded to undergraduate students with with the best academic performance by the Faculty of Engineering, Shanghai Jiao Tong University.
- Scholarship of Shanghai Jiao Tong University
2009 & 2010

Competitive institutional scholarship awarded to the top 10% undergraduate students with the best academic performance by Shanghai Jiao Tong University.
- The Third Prize of Applied Mathematics contest of Shanghai City
2008
- The First Prize of Physics contest of Shanghai City
2008

Entrepreneurship Awards (*change: WATER Labs*)

- MIT Water Innovation Prize Winner
2017

The first prize of MIT Water Innovation Competition, a startup competition focused on water innovation, with \$10k grant awarded by the MIT Water Club.
- MIT \$100k Entrepreneurship Competition Audience-Choice-Award
2017

The third prize of MIT \$100k Entrepreneurship Competition, one of the largest and world famous business plan competitions, with \$10k grant awarded by the MIT Entrepreneurship Center.
- Hult Prize@MIT Winner
2017

The first prize of Hult Prize Competition at MIT, one of the world famous social entrepreneurship awards.

- Harvard Arab Weekend 5th Start-Up Pitch Competition Winner
2016

The first prize of Harvard Arab Weekend 5th Start-Up Pitch Competition, with \$10k grant awarded by the Harvard Business School MENA club and the Harvard Arab Alumni Association.
- Tamer Social Impact Fund
2016

Winner of the seed grants of \$25k to nonprofit, for-profit, and hybrid early-stage social and environmental ventures awarded by Columbia University.
- MIT Enterprise Forum PanArab Innovation for Refugees Competition Winner
2016

The first prize of a global competition for the best tech-driven solutions addressing the challenges faced by refugees across the globe, with \$10k grant awarded by MIT Enterprise Forum PanArab Innovation for Refugees.
- Mass Clean Energy Council Catalyst Award
2016

Winner of the seed grants of \$50k to researchers and early-stage companies looking to demonstrate initial prototypes of their clean energy technologies awarded by the Mass Clean Energy Council.
- MIT IDEAS Global Challenge: Grant Winner
2015

The first prize of the MIT IDEAS Global Challenge competition, an annual innovation and social entrepreneurship competition for MIT students, with \$5K grant awarded by the MIT Innovation Initiative.